Prague University of Economics and Business
Faculty of Informatics and Statistics

A Decomposition of 10-K Risk-Factor Language and Post-Filing Stock Return Volatility

Bachelor Thesis

Study programme: Data Analytics

Author: Šimon Slanský
Supervisor: prof. RNDr. Ing. Michal Černý, Ph.D.

Prague, January 2027

Abstract

Corporate 10-K filings contain both audited financial data and narrative disclosures written by management, yet most empirical models use these two information channels separately. The Item 1A risk-factor section is itself the subject of two parallel literatures which predict effects in opposite directions: one, represented by Campbell et al. (2014), connects a larger *volume* of risk discussion to higher subsequent volatility, while the other, represented by Hope et al. (2016), links a higher *specificity* of risk discussion to lower volatility. This thesis proceeds in three steps. It first separates volume (the log word count of Item 1A) from specificity (the Loughran–McDonald risk-word density) and enters them jointly in a panel regression of post-filing log-volatility. It then splits the composite risk-density into its two sub-lists—uncertainty and litigious wording—each normalised by Item 1A length. Finally, it asks whether what a firm *omits* also matters, measuring for each firm-year the share of risk topics disclosed by industry peers but absent from its own Item 1A. All specifications use two-digit SIC industry and fiscal-year fixed effects with firm-clustered standard errors; the omission specification adds firm fixed effects to absorb permanent disclosure styles. On an unbalanced panel of 535 large U.S. firms over fiscal years 2010–2024 (5,691 firm-years at the 30-day horizon), longer Item 1A sections predict higher post-filing volatility ($\hat{\beta} = +0.130$, $t = +7.70$), while denser Item 1A sections, holding length constant, predict *lower* post-filing volatility ($\hat{\beta} = -3.88$, $t = -2.74$). The decomposition reveals that the entire negative effect is carried by the LM-Litigious sub-list ($\hat{\beta}^{\text{lit}} = -6.16$, $t = -3.35$), while the Uncertainty sub-list is statistically almost zero. The pattern survives at every horizon from 5 to 365 days and on both COVID-era and pre-COVID sub-samples. Consistent with this, the peer-relative omission gap is null in the cross-section but turns positive and significant under firm fixed effects ($\hat{\beta}^{\text{om}} = +1.345$, $t = +2.93$ at the 30-day horizon): firms that skip topics their peers do report have higher subsequent volatility. The channel operates only within-firm, once permanent section styles are absorbed.

Keywords

textual analysis, 10-K filings, SEC EDGAR, volatility, risk factors, Item 1A, Loughran–McDonald, LM-Litigious word list, boilerplate disclosure, panel data, corporate disclosures

Contents

List of Abbreviations

CIK Central Index Key
DV Dependent Variable
EDGAR Electronic Data Gathering, Analysis, and Retrieval
EPS Earnings Per Share
FE Fixed Effects
FY Fiscal Year
FYE Fiscal Year End
GLM Generalized Linear Model
L&M Loughran and McDonald
NLP Natural Language Processing
OCF Operating Cash Flow
OLS Ordinary Least Squares
ROA Return on Assets
SEC Securities and Exchange Commission
SIC Standard Industrial Classification
XBRL eXtensible Business Reporting Language

Introduction

1. Literature Review

2. Data

2.1 Selecting the Sample

We start with the sample of 1,000 largest U.S. public companies by market capitalisation, taken from the SEC EDGAR company-tickers file, and apply three filters. First, we keep only entities the SEC classifies as "operating," which removes exchange-traded funds, closed-end funds, blank-cheque companies, and shell entities. Second, we exclude firms in SIC codes 6000–6999 (finance, insurance, and real estate), following common sample-selection practice in prior accounting research (Francis et al., 2005; Sloan, 1996). Third, we require at least five years of readable XBRL data from 10-K filings and 90% valid stock-return coverage, so that there is enough within-firm variation for fixed-effects estimation.

Sample size. We deliberately focus on large, liquid firms rather than the full EDGAR universe for several reasons. First, our regressions use firm fixed effects with standard errors clustered by firm. As Petersen (2009) shows, the reliability of cluster-robust standard errors depends on the number of clusters—here 539—not on total observations, and 539 clusters are well above the threshold for reliable inference. Also, the textual-analysis part of the thesis requires downloading, parsing, and scoring every 10-K in the sample. Expanding to several thousand firms would greatly increase computation time. Lastly, smaller firms tend to have shorter XBRL histories (mandatory electronic filing was phased in by firm size), more tagging errors, and thinner trading, all of which add noise to financial ratios and return-based variables.

The sample size is in line with prior work combining textual and financial analysis of annual filings. Campbell et al. (2014) study about 9,000 firm-years of risk-factor disclosures from large U.S. firms; Loughran and McDonald (2011) use a broader set of roughly 50,000 firm-years, but their goal—validating a sentiment dictionary—calls for maximal cross-sectional coverage, whereas the joint volume–specificity test estimated here exploits within-industry-year variation across ~ 535 firms.

2.2 Choosing the Filing Frequency — why annual 10-Ks?

Before moving to variable construction, it is worth explaining why this study uses annual 10-K filings instead of quarterly 10-Q reports. Quarterly data would roughly triple the sample, but two practical problems—shown in Tables 2.1–2.2—ruled out that approach.

Financial data gaps in Q4. 10-Q filings cover only Q1–Q3, so any quarterly panel must reconstruct Q4 as the difference between the annual figure and the sum of the three preceding quarters. Table 2.1 shows that this residual approach is not usable for operating cash flow: the three quarterly facts coexist in only 7% of firm-years, because most firms report cumulative year-to-date cash flows rather than standalone quarterly amounts. Operating cash flow is the primary input into our OCF-to-assets ratio, so a quarterly panel would lose the cash-flow-based robustness check at the source. Even for net income and operating income, where coverage exceeds 80%, the implied Q4 disagrees with the explicit Q4 by more than one percent of the annual total in about 10–14% of comparable firm-years.

Table 2.1: Q4 Reliability: Availability and Consistency of Fourth-Quarter XBRL Facts

Metric	Firm-Years	Q1 – Q3 (%)	Explicit Q4 (%)	Comparable	Match (%)
Net Income	451	80.9	43.0	172	89.5
Operating Income	391	79.0	22.0	72	86.1
Operating Cash Flow	462	7.4	3.2	0	—

Note: "Q1 – Q3" indicates the share of fiscal years where all three quarterly facts (Q1, Q2, Q3) exist in XBRL. "Explicit Q4" indicates the share with a standalone fourth-quarter fact (period ≈90 days ending at fiscal year-end). "Comparable" is the count of firm-years for which both an explicit Q4 fact and the implied residual are available. "Match" reports the share of those comparable observations where the two values agree within 1% of the annual figure. Sample: 30 non-financial firms, all available fiscal years.

Missing risk-factor text in 10-Q. SEC Regulation S-K only requires firms to update risk factors in quarterly filings when "material changes" have occurred since the last 10-K. In practice, only about half the 10-Q filings in our pilot sample (15 firms, 90 filings) contained a legitimate risk-factor section. The rest had only a reference back to the annual report (Table 2.2). Constructing the year-on-year text variables that our specifications require is because of this not feasible at a quarterly frequency.

Chosen frequency. For these reasons—and following prior textual-analysis work based on 10-K filings (Li, 2008; Loughran & McDonald, 2011)—we use an annual panel. The XBRL coverage comparison (Table A.1) and a consolidated summary (Table A.2) are in Appendix A.

Table 2.2: Textual Data Availability: 10-K vs. 10-Q Filings

	10-K	10-Q
Filings tested	44	90
MD&A (Management's Discussion and Analysis)		
Section found (%)	93	100
Median word count	20,494	5,806
Risk Factors		
Substantive section found (%)	100	52
Boilerplate / reference only (%)	0	48
Median word count (substantive)	29,902	9,144

Note: "Found" requires >200 words extracted by regex-based section parser. "Substantive" for 10-Q excludes filings containing only a reference to the annual 10-K (e.g., "no material changes from those disclosed in our Annual Report"). MD&A corresponds to Item 7 in 10-K and Part I Item 2 in 10-Q; Risk Factors corresponds to Item 1A in 10-K and Part II Item 1A in 10-Q. Sample: 15 firms, most recent filings.

2.3 XBRL Tag Consistency Dilemma

A practical problem with the XBRL Company Facts API is that a single accounting concept can appear under different US-GAAP taxonomy tags. For example, operating cash flow may be reported as either `NetCashProvidedByUsedInOperatingActivities` (total, including discontinued operations) or `...ContinuingOperations` (continuing operations only). Net income may be tagged as `NetIncomeLoss`, `ProfitLoss`, or `IncomeLossFromContinuingOperations`. Stockholders' equity can include or exclude the non-controlling interest. These tags overlap a lot but are not identical: mixing them all within a single firm's time series introduces noise, while picking one tag globally would lose firms that consistently report under the other.

We handle this problem with a two-part *tag-harmonisation* process: (i) *equivalence groups* for tags that are economically interchangeable, and (ii) *per-firm tag locking* for tags that are not.

Equivalence groups. When two tags differ only in taxonomy version—not in their economic meaning—we let both enter a firm's time series. Specifically, the two operating-cash-flow tags and the pair `NetIncomeLoss`/`ProfitLoss` are treated as interchangeable. The third net-income variant, `IncomeLossFromContinuingOperations`, is *not* included because it deliberately excludes discontinued operations and can differ materially.

To verify this classification, we queried the SEC EDGAR Company Facts API for all 550 sample firms and identified 6,113 firm-year periods in which both tags of a pair were reported simultaneously; the overlap diagnostic supports treating each pair as interchangeable (median ROA bias 0.033 pp; details in Appendix A, Table A.4). The few cases with meaningful differences are handled by a priority rule that prefers the parent-company tag whenever both are reported.

Per-firm tag locking. For metrics where the alternative tags *are* economically distinct (stockholders' equity with vs. without NCI; `IncomeLossFromContinuingOperations` vs. `NetIncomeLoss`), we lock each firm to the tag it uses in the most fiscal years and set values from other tags to missing. Combined with the equivalence groups, this raises the complete-case rate for the five regression controls from 78.9% to 92.7% of firm-year observations. About 4% of firms end up with a continuing-operations OCF tag paired with a total net-income tag, a residual mismatch we accept and clip via winsorisation, in line with comparable XBRL studies (Chychyla & Kogan, 2015).

Per-firm tag choices and the overlap diagnostic are reported in Appendix A (Table A.4).

2.4 How do we extract text from 10-K filings?

This study heavily depends on the textual sections of 10-K reports Item 1A *Risk Factors* and Item 7 *Management's Discussion and Analysis of Financial Condition and Results of Operations* (MD&A). This section describes how we find and extract those sections from the raw HTML filings on SEC EDGAR.

2.4.1 Source of textual data

All filings come directly from the SEC EDGAR via the Submissions API, which links each company's Central Index Key (CIK) to the primary documents of its 10-K filings. From the panel of about 6,700 firm-year observations, we drop 10-K/A amendments. These typically consist of a cover page and a restated exhibit and do not contain the full narrative sections of the original 10-K making them unusable for further analysis. Financial data for amended fiscal years are kept, since the XBRL pipeline already picks up the amended figures when available (Section 2.3). Our extraction targets only original 10-K filings.

2.4.2 Section identification and exhibit fallback

Unlike structured XBRL data, the textual parts of 10-K filings are buried inside HTML with no standardised tagging, mixed with table-of-contents entries, running headers, and inline cross-references. We use a four-stage HTML parser that (i) detects candidate headers using regex patterns for the target "Item X" strings combined with formatting cues (bold, ALL-CAPS, $\geq 12\,\text{pt}$ font), rejecting hyperlinks and headers surrounded by other Item references within 300 characters (table-of-contents signature); (ii) tags confirmed headers in place before flattening to plain text; (iii) picks the candidate followed by the longest stretch of text before the next expected Item header; and (iv) bounds the section at the next Item header (Item 1B/1C/2 for Item 1A; Item 7A/8 for Item 7). If the recovered section is shorter than 200 words, the parser follows any reference pointing to an external exhibit (typically Exhibit 13) and reruns the same procedure with title-only patterns ("Risk Factors", "Management's Discussion and Analysis").

2.4.3 Post-Processing

After HTML-to-text conversion the parser strips three layout artefacts: "⟨*page number*⟩ *Table of Contents*" headers, standalone page numbers on lines that do not contain any other text, and non-breaking-space characters. The cleaning rules target only patterns that cannot be part of meaningful text and are applied after section boundaries have already been set.

2.4.4 Coverage and Limitations

The extraction pipeline was validated on two samples: the 50 most recent filings (fiscal year 2024, both sections recovered in all 50) and a stratified random sample of 100 filings covering 2010–2024 (97/100 successful, with three failures driven by PDF-only exhibits or the absence of any formatting cue for the section header). On the full panel we expect a coverage rate of roughly 97%, with missing values treated as missing in the textual variable construction.

LLM-assisted quality audit. To check that the extracted sections are actually the right ones, we ran an independent validation with a large language model (Google Gemini 2.5 Flash). A stratified random sample of 100 filings was drawn (20 per five-year period over 2010–2024) and both Item 1A and Item 7 were evaluated for every filing, giving 200 section-level judgments. The use of LLMs as independent validators in this role is supported by Gilardi et al. (2023); the sample size delivers Wilson 95 % confidence intervals of roughly ±3 percentage points (Agresti & Coull, 1998). The model graded each section on five criteria (correct section identity, start- and end-boundary accuracy, content quality, completeness) on a three-level scale (Pass / Minor / Fail). The detailed results are reported in Appendix A (Table B.1): on every individual criterion the lenient (Pass + Minor) pass rate is above 95 %, and the overall rate across all five criteria at once is 93.5 % (187/200; 95 % CI: 89.2–96.2 %). Failures cluster in older filings (5/40 in 2010–2012 vs. 1/40 in 2022–2024), reflecting the gradual standardisation of SEC filing HTML over the sample.

Relation to prior approaches. Our extraction strategy is inspired by Dyer et al. (2017), who document substantial evolution in 10-K disclosure—length, boilerplate, and structure—over time and the challenges this creates for large-sample studies. Compared with plain-text methods used in earlier work (Campbell et al., 2014; Li, 2008; Loughran & McDonald, 2011), our procedure avoids false positives from TOC entries and inline cross-references. Compared with XBRL-tagged textual blocks (e.g. `us-gaap:RiskFactorsTextBlock`), which became available with Inline XBRL adoption, our method covers a wider time span: structured textual tags are inconsistently filled before 2018, while the HTML-based approach works across the full 2010–2024 window.

3. Methodology

This chapter sets up the regressions that test H1, H2, and H3.

3.1 Outcome variable: post-filing realised volatility

The outcome is the natural log of annualised post-filing volatility, measured over a window of h calendar days that begins two trading days after the 10-K filing date:

$$\ln \sigma_{i,t+1}^{[h]} \;=\; \ln\!\left(\sqrt{252}\cdot \mathrm{sd}\big(r_{i,d}\big)_{d\in W_{i,t}^{[h]}}\right), \tag{3.1}$$

where $r_{i,d} = \ln(P_{i,d}/P_{i,d-1})$ is the daily log return for firm i and $W_{i,t}^{[h]}$ is the post-filing window for filing year t. The two-day lag avoids the same-day price reaction to the filing; the $\sqrt{252}$ factor annualises the daily standard deviation. The log transform makes the right-skewed distribution roughly symmetric and lets coefficients be read as percentage changes in volatility. The headline horizon is $h = 30$ days; results at $h \in \{5, 10, 90, 180, 365\}$ are reported as exploratory horizon checks.

3.2 Control variables

Every regression includes the same vector $\mathbf{X}_{i,t}^{\text{fin}}$ of firm-level controls drawn from the annual XBRL filings (Section 2.3): firm size ln(Total Assets), leverage (total liabilities over total assets), return on assets (net income over total assets), and asset growth (year-on-year change in total assets divided by lagged total assets). Lagged log-volatility $\ln \sigma_{i,t}^{[h]}$ enters with its own coefficient ϕ and is computed on the prior filing year's window of identical length h. All ratios are winsorised at the 1st and 99th percentiles of the regression sample. The H3 regression replaces asset growth with two liquidity proxies (current ratio and operating cash flow over assets). The reason is given in Section 3.5.

3.3 H1 — Item 1A length and disclosure volume

H1 follows the volume channel of Campbell et al. (2014): longer risk-factor sections signal more risk and should be followed by higher realised volatility. The disclosure-volume variable is the log word count of Item 1A:

$$\mathrm{Words}_{i,t}^{1A} \;=\; \text{number of tokens in Item 1A of firm } i\text{'s year-}t \text{ 10-K}, \tag{3.2}$$

and $\ln(\mathrm{Words}_{i,t}^{1A})$ enters the regression. The text-cleaning pipeline (Section 2.4.3) removes XBRL tags, footnote markers, and table cells before tokenisation, so the count reflects narrative content only. The median Item 1A holds about 8,800 words, with a long right tail (Section 4.1).

The volume channel is identified jointly with the specificity channel of H2 in a single regression (the *joint specification*):

$$\ln \sigma^{[h]}_{i,t+1} = \alpha + \phi \ln \sigma^{[h]}_{i,t} + \boldsymbol{\beta}'\mathbf{X}^{\text{fin}}_{i,t} + \beta^{\text{len}} \cdot \ln(\text{Words}^{1A}_{i,t}) + \beta^{\text{dens}} \cdot \text{RiskDensity}_{i,t} + \mu_j + \tau_t + \varepsilon_{i,t}, \quad (3.3)$$

where μ_j is a two-digit SIC industry fixed effect, τ_t is a fiscal-year fixed effect, and RiskDensity is defined in Section 3.4. H1 predicts $\beta^{\text{len}} > 0$. Entering length and density jointly is required because the dictionary risk-density used in prior work has length in its denominator, so the two channels are identified only if both are estimated together.

3.4 H2 — Risk-word density and disclosure specificity

H2 follows the specificity channel of Hope et al. (2016): holding length fixed, a section dominated by generic risk vocabulary carries less firm-specific information and should be followed by *lower* realised volatility. The test has two stages. The first stage uses a single composite risk-word density. The second stage splits that composite into its two Loughran and McDonald (2011) sub-lists to identify which one carries the effect.

3.4.1 Composite RiskDensity

The first-stage variable is the share of Item 1A words that match the LM uncertainty or litigious lists:

$$\text{RiskDensity}_{i,t} = \frac{n^{\text{unc}}_{i,t} + n^{\text{lit}}_{i,t}}{\text{Words}^{1A}_{i,t}}, \quad (3.4)$$

where n^{unc} and n^{lit} are the per-section counts of uncertainty- and litigious-list words. Higher values mean a larger share of the section consists of generic risk vocabulary. Lower values mean the narrative is dominated by firm-specific or non-dictionary content.

H2 is tested through the coefficient β^{dens} in the joint specification of equation 3.3, with the prediction $\beta^{\text{dens}} < 0$.

3.4.2 LM sub-list decomposition (second stage)

The composite RiskDensity is the sum of two sub-densities, each normalised by Item 1A length:

$$\text{UncDensity}_{i,t} = \frac{n^{\text{unc}}_{i,t}}{\text{Words}^{1A}_{i,t}}, \tag{3.5}$$

$$\text{LitDensity}_{i,t} = \frac{n^{\text{lit}}_{i,t}}{\text{Words}^{1A}_{i,t}}, \tag{3.6}$$

and RiskDensity = UncDensity + LitDensity. The two sub-lists capture different things: UncDensity picks up hedging language about what management does not know about its own business, while LitDensity picks up the cautious legal phrasing that lawyers add under the safe-harbour regime of Nelson and Pritchard (2007).

The second-stage regression replaces the composite with its two parts:

$$\begin{aligned} \ln \sigma^{[h]}_{i,t+1} &= \alpha + \phi \ln \sigma^{[h]}_{i,t} + \boldsymbol{\beta}' \mathbf{X}^{\text{fin}}_{i,t} + \beta^{\text{len}} \cdot \ln(\text{Words}^{1A}_{i,t}) \\ &\quad + \beta^{\text{unc}} \cdot \text{UncDensity}_{i,t} + \beta^{\text{lit}} \cdot \text{LitDensity}_{i,t} + \mu_j + \tau_t + \varepsilon_{i,t}. \end{aligned} \tag{3.7}$$

The two coefficients are separately identified because the sample correlation between UncDensity and LitDensity is close to zero ($r \approx -0.05$). The decomposition supports H2 if $\beta^{\text{lit}} < 0$ holds across horizons while β^{unc} is small and not significantly different from zero, indicating that the H2 effect comes from the litigious sub-list rather than from generic uncertainty hedging.

3.5 H3 — Peer-relative omission gap

H3 asks whether what a firm *leaves out* of Item 1A relative to its industry peers carries information about future volatility, beyond what length and density already capture.

Construction. For each fiscal year t we build a TF–IDF representation of every Item 1A in the panel, keeping terms that appear in at least five filings and at most 95% of filings, with ℓ_2-normalised rows. Let $\mathbf{x}_{i,t}$ be the TF–IDF row for firm i in year t. For each two-digit SIC industry j in year t with at least six panel filings, the leave-one-out peer aggregate is

$$\mathbf{p}_{i,t} = \sum_{k \in S_{j(i),t}, k \neq i} \mathbf{x}_{k,t}, \tag{3.8}$$

where $S_{j,t}$ is the set of panel firms in SIC2 industry j that filed in year t. The peer aggregate summarises what the focal firm's industry peers chose to write that year. The omission gap is the share of peer disclosure mass on terms the focal firm does not mention:

$$\textsc{Omission}^{1A}_{i,t} = 1 - \frac{\sum_{w: x_{i,t,w} > 0} p_{i,t,w}}{\sum_w p_{i,t,w}}, \tag{3.9}$$

with values in $[0, 1]$. Zero means the focal firm covers every term any peer mentions. Values near one mean the firm covers only a small share of what its peers chose to discuss. Firm-years in SIC2-year cells with fewer than five peers are dropped because the peer aggregate is too noisy.

Why SIC2 and why control for length. SIC2 is the standard peer benchmark in the disclosure-similarity literature (Brown & Tucker, 2011; Dyer et al., 2017). The median peer cell has 35 firms (P25 = 14). The omission gap is also negatively correlated with Item 1A length ($r = -0.85$): a longer section trivially covers more peer terms. The H3 regression therefore controls for $\ln(\text{Words}^{1A})$ so that the omission coefficient is read holding length fixed. The H3 control vector also replaces asset growth with two liquidity proxies, the current ratio and operating cash flow over assets.[1]

Specification. The H3 regression adds the omission gap to the joint specification:

$$\begin{aligned}\ln \sigma^{[h]}_{i,t+1} \;=\; \alpha \;+\; \phi \ln \sigma^{[h]}_{i,t} \;+\; \boldsymbol{\beta}'\mathbf{X}^{\text{fin}}_{i,t} \;+\; \beta^{\text{len}} \cdot \ln(\text{Words}^{1A}_{i,t}) \\ +\; \beta^{\text{dens}} \cdot \text{RiskDensity}_{i,t} \;+\; \beta^{\text{om}} \cdot \textsc{Omission}^{1A}_{i,t} \;+\; \mu_j \;+\; \tau_t \;+\; \varepsilon_{i,t}. \end{aligned} \tag{3.10}$$

Equation 3.10 is run under two fixed-effect structures: the cross-sectional version with μ_j a SIC2 industry fixed effect, and a stricter within-firm version in which μ_j is replaced by a firm fixed effect μ_i. The within-firm version is more demanding because it uses only year-to-year variation in omission driven by changes in peer disclosure. H3 is supported if $\beta^{\text{om}} > 0$ at the headline $h = 30$ days under both structures. Horizons run over $h \in \{5, 10, 30, 90, 180\}$ days. $h = 365$ days is dropped because the post-filing window for fiscal years 2023–2024 extends past the data cut-off.

The H3 sample is 4,630 to 4,636 firm-years, smaller than the H1+H2 sample of 5,691 at $h = 30$ days because of the ≥ 5 peers-per-cell restriction.

3.6 Estimation and inference

All regressions are estimated by ordinary least squares with the relevant fixed effects absorbed using the `linearmodels` package. Coefficients are identified from *within-industry, within-year* (or within-firm, within-year) variation, which removes both common time trends in volatility (the 2008 and 2020 crises in particular) and persistent industry- or firm-level differences in risk vocabulary. The analysis is predictive rather than causal: the coefficients measure the conditional association between Item 1A disclosure choices and *subsequent* realised volatility,

[1] The swap follows the risk-disclosure literature (Dyer et al., 2017; Kravet & Muslu, 2013), in which liquidity proxies are preferred to growth proxies because choices about what to omit from Item 1A are linked to short-run financing constraints rather than to asset-base expansion. Section 5.3 re-estimates equation 3.10 with the baseline H1+H2 control vector to confirm that the H3 result is not driven by the choice of controls.

in the spirit of Campbell et al. (2014), and should not be read as the effect of disclosure on volatility.

Primary horizon. The headline horizon is $h = 30$ days. The primary test of H1–H3 is conducted at this horizon. The other horizons in Table 5.1 are reported as exploratory evidence on the time decay of the signal.

Standard errors. Standard errors are two-way clustered by firm and fiscal year (Cameron et al., 2011; Petersen, 2009), because (i) consecutive firm-year volatility windows overlap, producing strong within-firm serial correlation, and (ii) calendar-year shocks (the 2020 COVID shock above all) produce strong within-year cross-sectional correlation that one-way firm clustering would miss.

Outliers. Financial ratios are winsorised at the 1st and 99th percentiles of the regression sample. The text variables are not winsorised: $\ln(\text{Words}^{1A})$ is bounded by the log transform and RiskDensity is a ratio with sample maximum ≈ 0.05.

Sample. The estimation sample at $h = 30$ days has 5,691 firm-years on 535 unique firms; the $h = 365$ days sample retains 6,172 firm-years on the same firms.

4. Results

This chapter reports the empirical results.

4.1 Descriptive statistics

Table 4.1 reports summary statistics for the 30-day regression sample. The dependent variable has mean $\ln \sigma_{i,t+1}^{[30]} \approx -1.34$, corresponding to a typical annualised post-filing volatility around 0.26. Median Item 1A length is roughly 8,800 words, and the LM uncertainty and litigious sub-shares are essentially uncorrelated ($r \approx -0.05$).

Table 4.1: Summary Statistics

	Mean	SD	P25	Median	P75
$\ln(\sigma_{i,t+1})$	-1.344	0.588	-1.740	-1.410	-1.058
$\ln(\sigma_{i,t})$	-1.353	0.590	-1.749	-1.424	-1.065
ln(Total Assets)	23.182	1.452	22.191	23.206	24.236
Leverage	0.616	0.218	0.476	0.611	0.739
ROA	0.063	0.087	0.027	0.062	0.106
Asset Growth	0.113	0.243	0.001	0.059	0.142
$\ln(\text{Words}^{1A})$	9.026	0.637	8.639	9.079	9.418
RiskDensity	0.051	0.006	0.047	0.051	0.055
UncDensity	0.034	0.005	0.032	0.034	0.037
LitDensity	0.017	0.005	0.013	0.016	0.019
Observations			5,691		
Firms			535		

Note: Summary statistics for the regression sample at the 30-day horizon. $\sigma_{i,t+1}$ is the annualised standard deviation of daily log returns over the 30-day window starting two trading days after the filing date; $\ln(\sigma_{i,t+1})$ is its natural logarithm. Words^{1A} is the cleaned word count of Item 1A; RiskDensity is the share of LM uncertainty- and litigious-list words in Item 1A; UncDensity and LitDensity are the two sub-components separately. Financial ratios are winsorised at the 1st and 99th percentiles.

Table 4.2 reports pairwise correlations. The two text variables are correlated at $r = -0.30$: longer Item 1A sections tend to have lower risk-word density, consistent with length being padded by firm-specific text rather than by extra generic risk vocabulary. The correlation is moderate enough to leave both signals separately identifiable in the joint specification.

Table 4.2: Pearson Correlation Matrix

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) $\ln(\sigma_{i,t+1})$	1							
(2) $\ln(\sigma_{i,t})$	0.35***	1						
(3) ln(Total Assets)	-0.19***	-0.20***	1					
(4) Leverage	-0.06***	-0.07***	0.21***	1				
(5) ROA	-0.23***	-0.25***	0.05***	-0.06***	1			
(6) Asset Growth	0.16***	0.09***	-0.12***	-0.12***	-0.04***	1		
(7) $\ln(\text{Words}^{1A})$	0.31***	0.31***	-0.09***	-0.04***	-0.27***	0.17***	1	
(8) RiskDensity	-0.11***	-0.12***	-0.01	-0.01	0.06***	-0.08***	-0.30***	1
(9) UncDensity	-0.09***	-0.08***	-0.18***	-0.06***	0.14***	-0.05***	-0.43***	0.71***
(10) LitDensity	-0.06***	-0.08***	0.18***	0.04***	-0.06***	-0.06***	0.03**	0.66***

Note: Pairwise Pearson correlation coefficients on the 30-day-horizon regression sample. ***, **, * denote significance at the 1%, 5%, and 10% levels.

4.2 Volume, specificity, and the LM-list decomposition

Table 4.3 reports five nested specifications at the 30-day horizon: financial baseline (1), length only (2), density only (3), the joint H1+H2 specification (4), and the LM-list decomposition (5).

In the joint specification (Column 4), longer Item 1A sections predict higher post-filing volatility ($\hat{\beta}^{\text{len}} = +0.121$, $t = +7.64$) and, holding length fixed, denser sections predict lower volatility ($\hat{\beta}^{\text{risk}} = -3.88$, $t = -2.68$). Both predictions hold in the same regression. The density coefficient halves between Columns 3 and 4: about half of what looked like a "risk-density effect" in the dictionary-only specification was in fact the volume effect operating through the section's word-count denominator. **H1 and H2 are both supported.**

Replacing the composite RiskDensity by its two LM sub-densities (Column 5) shows that the negative specificity association is carried almost entirely by the LM-Litigious sub-list: $\hat{\beta}^{\text{lit}} = -6.16$ ($t = -3.38$, $p < 0.001$), while $\hat{\beta}^{\text{unc}} = -1.28$ is not statistically distinguishable from zero. A Wald test of equality yields $z = +1.91$ ($p = 0.056$), borderline at the 5% level. The pattern is consistent with Beatty et al. (2019), Cazier et al. (2021), and Nelson and Pritchard (2007): litigious-list language is standardised cautionary boilerplate, and once it is isolated from the uncertainty sub-list it does most of the H2 work. Adjusted R^2 rises only modestly across the five columns ($0.492 \rightarrow 0.507$) because the financial baseline already captures the bulk of predictable volatility. Table 4.4 reproduces the same five columns with the dependent variable replaced by $\ln \sigma^{[365]}_{i,t+1}$; the signs are preserved and magnitudes shrink to roughly half of the 30-day estimates.

Table 4.3: Item 1A Length, Risk-Word Density, and the Boilerplate Decomposition (30 days window)

	(1)	(2)	(3)	(4)	(5)
$\ln(\sigma_{i,t})$	+0.1695**	+0.1368*	+0.1552**	+0.1336*	+0.1321*
	(+2.12)	(+1.80)	(+1.96)	(+1.76)	(+1.75)
ln(Total Assets)	−0.0700***	−0.0671***	−0.0695***	−0.0672***	−0.0645***
	(−5.81)	(−5.91)	(−5.99)	(−5.98)	(−5.69)
Leverage	−0.0198	−0.0026	−0.0249	−0.0072	−0.0082
	(−0.56)	(−0.07)	(−0.72)	(−0.21)	(−0.24)
ROA	−1.0678***	−0.8982***	−1.0689***	−0.9189***	−0.9250***
	(−9.72)	(−8.68)	(−9.80)	(−8.80)	(−8.77)
Asset Growth	+0.2460***	+0.2033***	+0.2313***	+0.2011***	+0.1982***
	(+4.34)	(+3.80)	(+4.21)	(+3.78)	(+3.72)
$\ln(\text{Words}^{1A})$		+0.1375***		+0.1212***	+0.1298***
		(+8.43)		(+7.64)	(+7.16)
RiskDensity			−7.7820***	−3.8769***	
			(−5.40)	(−2.68)	
UncDensity					−1.2779
					(−0.62)
LitDensity					−6.1591***
					(−3.38)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	5,691	5,691	5,691	5,691	5,691
Firms	535	535	535	535	535
Adj. R^2	0.492	0.505	0.497	0.506	0.507

Note: The dependent variable is $\ln(\sigma_{i,t+1})$, the natural log of annualised post-filing volatility over a 30 days window starting two trading days after the 10-K filing date. Column (1) reports the financial baseline; column (2) adds the log word count of Item 1A (H1); column (3) adds the composite LM risk-word density (H2); column (4) adds H1 and H2 jointly; column (5) replaces the H2 composite by its LM-list decomposition into the LM-Uncertainty and LM-Litigious densities of Item 1A. All specifications include two-digit SIC industry and fiscal-year fixed effects. t-statistics in parentheses use two-way (firm and fiscal-year) clustered standard errors. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Table 4.4: Item 1A Length, Risk-Word Density, and the Boilerplate Decomposition (365 days window)

	(1)	(2)	(3)	(4)	(5)
$\ln(\sigma_{i,t})$	+0.6112***	+0.5731***	+0.5935***	+0.5680***	+0.5657***
	(+14.37)	(+12.60)	(+13.41)	(+12.34)	(+12.25)
ln(Total Assets)	−0.0339***	−0.0346***	−0.0346***	−0.0349***	−0.0337***
	(−4.76)	(−4.94)	(−4.91)	(−5.00)	(−4.76)
Leverage	−0.0116	−0.0026	−0.0139	−0.0051	−0.0056
	(−0.81)	(−0.17)	(−0.94)	(−0.34)	(−0.37)
ROA	−0.4402***	−0.3901***	−0.4552***	−0.4054***	−0.4101***
	(−7.45)	(−7.20)	(−7.54)	(−7.43)	(−7.42)
Asset Growth	+0.0728**	+0.0586**	+0.0684**	+0.0579**	+0.0568**
	(+2.49)	(+2.29)	(+2.38)	(+2.25)	(+2.20)
$\ln(\text{Words}^{1A})$		+0.0645***		+0.0558***	+0.0607***
		(+5.15)		(+4.53)	(+4.93)
RiskDensity			−3.8105***	−2.1913***	
			(−5.42)	(−3.55)	
UncDensity					−0.8552
					(−1.08)
LitDensity					−3.3905***
					(−3.91)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	6,172	6,172	6,172	6,172	6,172
Firms	535	535	535	535	535
Adj. R^2	0.729	0.734	0.731	0.735	0.735

Note: The dependent variable is $\ln(\sigma_{i,t+1})$, the natural log of annualised post-filing volatility over a 365 days window starting two trading days after the 10-K filing date. Column (1) reports the financial baseline; column (2) adds the log word count of Item 1A (H1); column (3) adds the composite LM risk-word density (H2); column (4) adds H1 and H2 jointly; column (5) replaces the H2 composite by its LM-list decomposition into the LM-Uncertainty and LM-Litigious densities of Item 1A. All specifications include two-digit SIC industry and fiscal-year fixed effects. t-statistics in parentheses use two-way (firm and fiscal-year) clustered standard errors. ***, **, * denote significance at the 1%, 5%, and 10% levels.

4.3 Strategic silence and within-firm post-filing volatility

Table 4.5 reports the H3 specification of equation 3.10 at five post-filing horizons under two fixed-effect structures: industry plus year FE in Panel A and firm plus year FE in Panel B. The coefficient of interest is β^{om} on the peer-relative omission gap; Item 1A length, RiskDensity, the five financial controls, and lagged log-volatility are partialled out.

Table 4.5: Silence Hypothesis: Item 1A omission relative to SIC2 peers and post-filing realised volatility.

	$h = 5$	$h = 10$	$h = 30$	$h = 90$	$h = 180$
Panel A. Industry × year fixed effects					
OMISSION$_{1A}$	+0.424	-0.054	+0.098	+0.179	+0.226
	(+0.71)	(-0.11)	(+0.28)	(+0.63)	(+1.04)
N	4,630	4,603	4,636	4,636	4,636
R^2	0.174	0.354	0.505	0.575	0.650
Panel B. Firm × year fixed effects					
OMISSION$_{1A}$	+0.808	+1.233	+1.345***	+1.299**	+1.136**
	(+0.81)	(+1.50)	(+2.93)	(+2.42)	(+2.35)
N	4,630	4,603	4,636	4,636	4,636
R^2	0.322	0.484	0.627	0.691	0.750

Notes. Dependent variable is $\log(\text{RV}_{hd})$ over the h-day window beginning the trading day after filing. OMISSION$_{1A}$ is the share of TF-IDF mass on terms used by SIC2-year peers (leave-one-out, ≥ 5 peers) that the focal firm does not mention. All regressions control for $\log(\text{RV}_{hd,t-1})$, the five financial controls (size, leverage, ROA, current ratio, OCF/assets), $\log(\text{words}_{1A})$, and RISKDENSITY$_{1A}$. t-statistics in parentheses use two-way clustered standard errors on (ticker, fiscal year). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The cross-section in Panel A is null at every horizon ($|t| < 1.1$). In the firm-FE specification of Panel B the coefficient turns positive and significant at the 30-day horizon ($\hat{\beta}^{\text{om}} = +1.345$, $t = +2.93$, $p = 0.003$) and survives at 90 and 180 days. A one-within-firm-SD rise in OMISSION^{1A} (0.054) implies a $\approx 7.3\%$ increase in 30-day annualised volatility, of the same order as the H1 length effect. The reading is that permanent firm-level disclosure styles absorb the cross-section, while the peer-relative omission gap—which by construction varies with what *other* firms write each year—survives the firm-FE filter. **H3 is supported within-firm at the 30-day horizon**, with the placebo, length-orthogonalisation, large-cell, decile-rank, and lead-lag qualifications discussed in Section 5.3.

5. Robustness

This chapter checks that the headline results of Chapter 4 survive plausible variations in the specification. Section 5.1 re-estimates the H1+H2 decomposition specification at five additional post-filing horizons. Section 5.2 re-estimates the same specification on the pre-COVID and COVID-era sub-samples. Section 5.3 replaces industry fixed effects with firm fixed effects. Section 5.3 reports the placebo, length-residualised, large-cell, decile-rank, and prior-year specifications of the H3 strategic-silence result. Section 5.4 closes with a brief note on alternative specificity-channel candidates that were tested and rejected before settling on the LM-list decomposition.

5.1 Alternative post-filing windows

The headline specification measures post-filing volatility over a 30-day window. To check the choice of window we re-estimate the H1+H2 specification of equation 3.7 at five additional horizons (5, 10, 90, 180, and 365 days). Table 5.1 reports all six columns side by side.

The pattern is consistent in three respects. First, all three text coefficients keep the predicted sign at every horizon—positive for $\ln(\text{Words}^{1A})$, negative for LitDensity, and statistically indistinguishable from zero for UncDensity—so neither H1 nor the LM-list decomposition reverses for any plausible choice of window. Second, the LitDensity coefficient is significant at the 1% level from 10 days onward, with $|t|$ between 2.88 (10 days) and 4.02 (180 days). Third, the UncDensity coefficient never approaches statistical significance, with $|t| \leq 1.1$ at every window. The 30-day choice for the headline specification is therefore not a window in which the results happen to be sharpest—they are sharp at every horizon the data admit.

Both coefficient series also decline approximately monotonically in h, the empirical signature of an event-driven signal that is incorporated into prices over weeks rather than years (Loughran & McDonald, 2011; Tetlock, 2007). No formal half-life is fitted because the across-horizon error terms are correlated through the shared underlying daily returns.

Table 5.1: Post-Filing Volatility at Alternative Horizons

	(1) 5d	(2) 10d	(3) 30d	(4) 90d	(5) 180d	(6) 365d
$\ln(\sigma_{i,t})$	+0.1008***	+0.1249***	+0.1321*	+0.2122**	+0.3533***	+0.5657***
	(+3.62)	(+4.55)	(+1.75)	(+2.24)	(+4.19)	(+12.25)
$\ln(\text{Words}^{1\text{A}})$	+0.1953***	+0.1524***	+0.1298***	+0.1155***	+0.0962***	+0.0607***
	(+8.78)	(+6.72)	(+7.16)	(+5.93)	(+5.11)	(+4.93)
UncDensity	+0.9530	−2.1555	−1.2779	−0.7636	−0.6946	−0.8552
	(+0.27)	(−1.04)	(−0.62)	(−0.45)	(−0.55)	(−1.08)
LitDensity	−7.0957***	−6.6115***	−6.1591***	−6.5495***	−5.1636***	−3.3905***
	(−3.76)	(−2.88)	(−3.38)	(−3.94)	(−4.02)	(−3.91)
Financial controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,690	5,648	5,691	5,691	5,691	6,172
Firms	535	535	535	535	535	535
Adj. R^2	0.156	0.354	0.507	0.575	0.643	0.735

Note: Each column re-estimates the H1 + LM-list decomposition specification replacing the dependent variable with $\ln(\sigma^{[h]}_{i,t+1})$ measured over a post-filing window of length h. The lagged dependent variable is the same firm's log-volatility computed over the prior filing year's window of identical length. UncDensity and LitDensity are the LM-Uncertainty and LM-Litigious word densities on Item 1A; together they sum to RiskDensity (H2). Financial controls (size, leverage, ROA, asset growth) are included but suppressed. Industry and year fixed effects are absorbed; t-statistics in parentheses use two-way (firm and fiscal-year) clustered standard errors. ***, **, * denote significance at the 1%, 5%, and 10% levels.

5.2 Pre-COVID and COVID-era sub-samples

The 2020 fiscal year and its aftermath were a structural shock for both equity volatility and corporate disclosure: average annualised volatility roughly doubled in the spring of 2020 and Item 1A sections grew systematically longer as firms added pandemic and supply-chain language. To check that the headline results are not driven by either the pre-COVID baseline or the COVID-era spike, we re-estimate the H1+H2 decomposition specification (equation 3.7) separately on the two sub-periods. Filings with fiscal year $\leq$ 2019 form the pre-COVID sample (3,243 firm-years) and filings with fiscal year $\geq$ 2020 form the COVID-era sample (2,448 firm-years). All other features of the specification are unchanged. Table 5.2 reports the result.

Table 5.2: Sub-Period Robustness (30-day horizon)

	(1) Pre-COVID (2010–2019)	(2) COVID era (2020–2024)
$\ln(\sigma_{i,t})$	+0.1271	+0.1313*
	(+1.21)	(+1.72)
$\ln(\text{Words}^{1A})$	+0.1025***	+0.1715***
	(+6.06)	(+8.92)
UncDensity	−2.8830	+0.8033
	(−1.17)	(+0.33)
LitDensity	−5.4863**	−9.8819***
	(−2.55)	(−4.85)
Financial controls	Yes	Yes
Industry FE	Yes	Yes
Year FE	Yes	Yes
Observations	3,243	2,448
Firms	470	527
Adj. R^2	0.529	0.426

Note: H1 + LM-list decomposition specification re-estimated on two non-overlapping sub-samples. The dependent variable is $\ln(\sigma^{[30]}_{i,t+1})$. UncDensity and LitDensity are the LM-Uncertainty and LM-Litigious word densities on Item 1A. All financial controls and SIC2+year fixed effects are included. Standard errors are two-way clustered by firm and fiscal year. ***, **, * denote significance at the 1%, 5%, and 10% levels.

The two active text channels are present in both sub-periods with the predicted signs and at the same order of magnitude. Length is significant at the 1% level in both ($\beta = +0.103$, $t = +6.06$ pre-COVID; $\beta = +0.172$, $t = +8.92$ COVID era). LitDensity is negative and significant in both ($\beta = -5.49$, $t = -2.55$ pre-COVID; $\beta = -9.88$, $t = -4.85$ COVID era). UncDensity is statistically indistinguishable from zero in both samples, consistent with the headline finding that uncertainty wording adds no explanatory power once length and litigious wording are controlled for. The two active coefficients (length and LitDensity) are larger in absolute value in the COVID-era sample, consistent with the higher overall level of volatility in that period giving more scale to the disclosure signal, but the qualitative pattern is identical. The headline finding is not a COVID artefact.

5.3 Within-firm identification: firm fixed effects

The headline specification absorbs two-digit SIC industry and fiscal-year fixed effects. Coefficients are therefore identified from variation *within an industry-year cell*, which removes the most plausible confounders (industry-level differences in risk vocabulary and macroeconomic time effects) but still allows permanent firm-level traits to drive the result. To rule that out we replace the industry fixed effects with firm fixed effects, so that identification comes only from *within-firm* year-to-year variation in the three text variables. Table 5.3 reports the result at the 30-day horizon.

Table 5.3: Within-Firm Identification (Firm + Year FE, 30-day horizon)

	Estimate
$\ln(\sigma_{i,t})$	-0.1059***
	(-6.77)
$\ln(\text{Words}^{1A})$	$+0.0835$***
	$(+3.31)$
UncDensity	-1.4559
	(-0.55)
LitDensity	$+0.9561$
	$(+0.35)$
Financial controls	Yes
Firm FE	Yes
Year FE	Yes
Observations	5,691
Firms	535
Adj. R^2	0.583

Note: H1 + LM-list decomposition specification with firm and fiscal-year fixed effects absorbed. Identification of $\ln(\text{Words}^{1A})$, UncDensity and LitDensity comes only from within-firm variation across years. Standard errors clustered by firm. ***, **, * denote significance at the 1%, 5%, and 10% levels.

The volume channel survives: $\hat{\beta}^{\text{len}} = +0.084$ ($t = +3.31$, $p < 0.001$). Firms that grow their Item 1A from year to year do experience higher subsequent volatility. Both LM sub-densities, by contrast, are absorbed: $\hat{\beta}^{\text{unc}} = -1.46$ ($t = -0.55$) and $\hat{\beta}^{\text{lit}} = +0.96$ ($t = +0.35$). Risk-vocabulary composition has very little within-firm variation across years because Item 1A is largely recycled from one filing to the next (Cazier et al., 2021), so the firm-FE specification has almost no remaining variance to identify a sub-density coefficient from. This is consistent with the Beatty et al. (2019), Cazier et al. (2021), and Nelson and Pritchard (2007) reading of cautionary boilerplate as a quasi-permanent firm-level disclosure-style trait. The headline H2 result is therefore read as a between-firm regularity, without claiming a within-firm causal channel.

The headline H3 result of Section 4.3 relies on a within-firm specification at the 30-day horizon. Three features of the construction make further robustness checks essential before reading the result as evidence for strategic silence rather than for some confounding artefact: (i) the omission gap is mechanically negatively correlated with Item 1A length ($r = -0.85$), so any insufficiently strong control for length would let omission proxy for length; (ii) the SIC2 peer benchmark is one of several plausible reference sets, and the result might be specific to that choice; (iii) the contemporaneous design has a partial overlap between the filing date and the start of the post-filing volatility window, leaving some scope for reverse causality if a difficult year produces both a shorter, less peer-aligned 10-K and higher subsequent volatility for unrelated reasons. Table 5.4 reports six robustness specifications designed to address these concerns; a placebo test is reported separately below.

Table 5.4: Robustness of the strategic-silence result (H3): firm $\times$ year fixed effects, $h = 30$ days.

	(1) Baseline	(2) Decile rank	(3) Length-resid	(4) ≥ 20 peers	(5) Triple length	(6) Prior-year
OMISSION_{1A}	+1.345***	+0.010	+0.352	+2.519***	+1.502***	+0.485**
	(+2.93)	(+1.31)	(+1.49)	(+3.09)	(+3.18)	(+1.97)
N	4,636	3,706	4,636	3,248	4,636	4,546

Notes. Each column is a separate firm $\times$ year fixed-effect regression at $h = 30$ days. Column (1) reports the baseline H3 specification of equation 3.10. Column (2) replaces OMISSION_{1A} with its decile rank within (SIC2, fiscal year) cells. Column (3) replaces it with the residual after regressing OMISSION_{1A} on $\ln(\text{WORDS}_{1A})$ within (SIC2, fiscal year) cells. Column (4) restricts the sample to industry-year cells with at least 20 panel firms. Column (5) adds $\ln(\text{WORDS}_{\text{MDA}})$ and $\ln(\text{WORDS}_{1A})^2$ as additional length controls. Column (6) replaces the contemporaneous omission with its prior-year value. All regressions include lagged log-volatility, the five financial controls, $\ln(\text{WORDS}_{1A})$, and RISKDENSITY_{1A}. t-statistics in parentheses use two-way clustered standard errors on (ticker, fiscal year). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Placebo (peer-cell shuffle). The first and most important robustness check holds the SIC2 cell sizes and the firm-year distribution constant but randomly permutes which firm belongs to which SIC2 cell within fiscal year, then recomputes the omission gap of equation 3.9 and re-estimates equation 3.10 with firm and year fixed effects. If the omission–volatility association reflects something specifically about being silent on *peer-relevant* risks, the placebo coefficient should die. If the result is a length artefact, the placebo should mirror the real coefficient. Across five independent permutations the placebo coefficient is small and statistically insignificant in every draw, with $|t|$ values $0.88, 0.90, 0.54, 0.57, 0.63$. The real coefficient $t = +2.93$ exceeds the maximum placebo $|t|$ by a factor of three. The strategic-silence reading of the H3 result therefore survives a clean placebo test on the peer-structure mechanism.

Length is not the driver (Columns 3 and 5). Column (3) replaces the omission gap by its residual after regressing on $\ln(\textsc{Words}_{1A})$ within each (SIC2, fiscal year) cell; the purified variation enters with the right sign but loses significance ($t = +1.49$). Column (5) keeps the original gap but adds two further length controls (MD&A word count and squared Item 1A log word count); the omission coefficient strengthens to $t = +3.18$ ($p = 0.001$). Linear or quadratic length controls do not remove the signal, but the strict orthogonalisation of Column (3) absorbs enough of the effect to leave a borderline rather than a strong association.

Large-cell restriction (Column 4) and prior-year specification (Column 6). Restricting to cells with at least 20 peers reduces the panel to 3,248 firm-years but *strengthens* the coefficient to +2.52 ($t = +3.09$): the signal is sharper where the peer aggregate is more reliably estimated. Column (6) replaces the contemporaneous omission with its prior-year value to address the filing-window overlap concern; the lagged coefficient remains positive and significant at the 5% level (+0.485, $t = +1.97$).

Decile rank (Column 2). Replacing the level of omission by its decile rank within each (SIC2, fiscal year) cell preserves the predicted positive sign but reduces significance to $t = +1.31$. The H3 effect is concentrated in the tails of the omission distribution, consistent with a strategic-silence reading in which conspicuous omissions drive the response. The decile-rank weakness is the H3 result's most important qualification.

Bottom line on H3. The strategic-silence result survives the placebo, two stronger length controls, the large-cell restriction, and the one-year-lag specification, with the right sign in every column. Two qualifications apply: the decile-rank specification is not significant at the 5% level, and the fully length-orthogonalised reformulation (Column 3) is borderline rather than significant. The thesis reads the H3 evidence as supportive of a within-firm strategic-silence channel with these stated qualifications, and does not extend the claim to a causal interpretation.

5.4 What did not work

Seven alternative specificity-channel candidates were estimated on the same panel and rejected before the LM-list decomposition was adopted: (i) year-on-year TF–IDF similarity of Item 1A in the spirit of Brown and Tucker (2011); (ii) MD&A LM-Negativity; (iii) the first difference $\Delta \ln(\text{Words}^{1A})$ following Lyle et al. (2023); and (iv)–(vii) four functional forms of FinBERT-tone (`yiyanghkust/finbert-tone`) aggregated from the per-sentence probabilities ($P(\text{neg})$, the signed $P(\text{pos}) - P(\text{neg})$, $P(\text{neutral})$, and the intensity $1 - P(\text{neutral})$). Each candidate was added as a third text variable to the H1+H2 joint specification of equation 3.3; none added statistically significant explanatory power for $\ln \sigma^{[30]}_{i,t+1}$ ($|t| < 1.5$ in every case).

The TF–IDF and FinBERT signals capture firm-specific narrative changes whose informational content is, on this panel, dominated by the much stronger length signal of H1. The LM-list decomposition succeeds where these alternatives fail because it identifies a structural composition of the H2 specificity effect rather than an additional firm-specific signal that competes with H1.

Conclusion

References

Agresti, A., & Coull, B. A. (1998). Approximate is better than "exact" for interval estimation of binomial proportions. *The American Statistician*, *52*(2), 119–126.

Beatty, A., Cheng, L., & Zhang, H. (2019). Are risk factor disclosures still relevant? evidence from market reactions to risk factor disclosures before and after the financial crisis. *Contemporary Accounting Research*, *36*(2), 805–838.

Brown, S. V., & Tucker, J. W. (2011). Large-sample evidence on firms' year-over-year MD&A modifications. *Journal of Accounting Research*, *49*(2), 309–346.

Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics*, *29*(2), 238–249.

Campbell, J. L., Chen, H., Dhaliwal, D. S., Lu, H.-m., & Steele, L. B. (2014). The information content of mandatory risk factor disclosures in corporate filings. *Review of Accounting Studies*, *19*(1), 396–455.

Cazier, R. A., McMullin, J. L., & Treu, J. S. (2021). Are lengthy and boilerplate risk factor disclosures inadequate? an examination of judicial and regulatory assessments of risk factor language. *The Accounting Review*, *96*(4), 131–155.

Chychyla, R., & Kogan, A. (2015). Using XBRL to conduct a large-scale study of discrepancies between the accounting numbers in Compustat and SEC 10-K filings. *Journal of Information Systems*, *29*(1), 37–72. https://doi.org/10.2308/isys-50922

Dyer, T., Lang, M., & Stice-Lawrence, L. (2017). The evolution of 10-K textual disclosure: Evidence from Latent Dirichlet Allocation. *Journal of Accounting and Economics*, *64*(2–3), 221–245.

Francis, J., LaFond, R., Olsson, P. M., & Schipper, K. (2005). The market pricing of accruals quality. *Journal of Accounting and Economics*, *39*(2), 295–327.

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd-workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, *120*(30), e2305016120.

Hope, O.-K., Hu, D., & Lu, H. (2016). The benefits of specific risk-factor disclosures. *Review of Accounting Studies*, *21*(4), 1005–1045.

Kravet, T., & Muslu, V. (2013). Textual risk disclosures and investors' risk perceptions. *Review of Accounting Studies*, *18*(4), 1088–1122.

Li, F. (2008). Annual report readability, current earnings, and earnings persistence. *Journal of Accounting and Economics*, *45*(2–3), 221–247.

Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance*, *66*(1), 35–65.

Lyle, M. R., Riedl, E. J., & Siano, F. (2023). Changes in risk factor disclosures and the variance risk premium. *The Accounting Review*, *98*(5), 327–360.

Nelson, K. K., & Pritchard, A. C. (2007). Litigation risk and voluntary disclosure: The use of meaningful cautionary language. *Working Paper, University of Michigan / Rice University.*

Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *The Review of Financial Studies*, *22*(1), 435–480.

Sloan, R. G. (1996). Do stock prices fully reflect information in accruals and cash flows about future earnings? *The Accounting Review*, *71*(3), 289–315.

Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *The Journal of Finance*, *62*(3), 1139–1168.

Appendices

A. Variable Definitions and Data Pipeline

This appendix collects supplementary tables referenced in the main text. Section A.1 presents the diagnostics behind the choice of annual filing frequency (Section 2.2). Section A.2 reports the per-firm XBRL tag assignments from the tag-harmonisation procedure (Section 2.3).

A.1 Filing Frequency Diagnostics

The following tables supplement the discussion in Section 2.2.

Table A.1: XBRL Financial Metric Coverage: Annual (10-K) vs. Quarterly (10-Q)

Metric	Annual (%)	Quarterly (%)
assets_current	87.0	91.1
eps_diluted	89.4	92.7
liabilities_current	87.0	91.1
net_income	93.6	97.3
operating_cash_flow	89.2	35.8
operating_income	76.9	80.5
stockholders_equity	98.7	94.9
total_assets	89.9	93.3
total_liabilities	63.7	66.5
Median periods per firm	20	52
Firms with data	30	30

Note: Coverage shows the percentage of firm-periods with non-missing values for each metric. Annual data is extracted from 10-K filings; quarterly data from 10-Q filings (Q1–Q3 only). Sample: 30 non-financial operating companies from the top 200 by market capitalisation.

Table A.2: Annual vs. Quarterly Approach: Summary Comparison

Criterion	Annual (10-K)	Quarterly (10-Q)
Filing type	10-K	10-Q (Q1–Q3 only)
MD&A availability	93% (Item 7)	100% (Part I, Item 2)
Risk Factors availability	100% (Item 1A)	∼52% substantive
Risk Factors content	Median ∼30,000 words	Median ∼9,000 words (substantive); 48% boilerplate
Q4 financial data	Included in 10-K	Unreliable / absent
Explicit Q4 XBRL facts	N/A	3–43% of firm-years
Implied Q4 mismatch rate	N/A	∼11% of comparisons
Cash-flow quarterly coverage	N/A	7% of firm-years
Item 1A length (H1)	Feasible (annual word count)	Limited (52% substantive)
Item 1A risk-word density (H2)	Feasible (annual LM ratio)	Limited (52% substantive)
Literature alignment	Standard	Uncommon
Expected observations	∼6,400 firm-years	∼19,000 firm-quarters

Note: Summary based on diagnostic tests of 30 non-financial companies (XBRL, Q4 tests) and 15 companies (textual tests). Risk Factors in 10-Q filings are required by SEC Regulation S-K Item 1A only when there are "material changes" from the most recent 10-K; roughly half of companies include substantive risk disclosures, while the rest provide only a boilerplate reference. See Tables A.1, 2.1, and 2.2 for detailed results.

Table A.3: Per-Company 10-Q Textual Content (Most Recent 6 Filings)

Ticker	Filings	MD&A Found	MD&A Med. Words	Risk Subst.	Risk Boilerplate
AAPL	6	6	2,781	3	3
AMZN	6	6	6,291	6	0
AVGO	6	6	4,778	6	0
COST	6	6	4,979	0	6
GOOGL	6	6	8,096	3	3
JNJ	6	6	7,587	0	6
LLY	6	6	4,994	0	6
MA	6	6	7,300	6	0
META	6	6	7,756	6	0
MSFT	6	6	7,981	0	6
NVDA	6	6	5,156	6	0
TSLA	6	6	5,582	2	4
V	6	6	5,043	0	6
WMT	6	6	7,142	3	3
XOM	6	6	6,197	6	0

Note: "MD&A Found" and "Risk Subst." count filings with >200 substantive words. "Risk Boilerplate" counts filings where the risk-factors section contains only a reference to the annual 10-K.

A.2 XBRL Tag Provenance

Table A.4 reports, for each metric with multiple candidate XBRL tags, the number and share of firms locked to each tag by the per-firm tag-locking procedure described in Section 2.3.

Table A.4: Per-firm XBRL tag distribution

Metric	Locked XBRL tag	Firms	%
Net income	`NetIncomeLoss`	502	91.6
	`ProfitLoss`	45	8.2
	`IncomeLossFromContinuingOperations`	1	0.2
Operating cash flow	`NetCashProvidedByUsedInOperatingActivities`	530	96.7
	`...OperatingActivitiesContinuingOperations`	18	3.3
Stockholders' equity	`StockholdersEquityIncl...NoncontrollingInterest`	318	58.6
	`StockholdersEquity`	225	41.4

Notes. For each (firm, metric) pair, the tag with the highest within-firm year-count is selected; values from other tags are set to missing. Firm counts are based on up to 548 firms in the tag-locking stage; the stockholders' equity row sums to 543 because five firms report no equity tag at all. See Section 2.3 for the rationale.

B. Documentation of AI Tool Usage

This appendix documents the use of artificial intelligence tools in preparing this thesis, as required by the university.

Extraction Validation (Section 2.4.4)

Tool. Google Gemini 2.5 Flash (accessed via the OpenAI-compatible Gemini API endpoint), with temperature set to 0 for deterministic output.

Purpose. An independent, automated quality audit of the SEC 10-K section extraction pipeline. The model evaluated 200 extracted sections (100 Item 1A, 100 Item 7) against a structured five-criterion rubric (correct section, start boundary, end boundary, content quality, completeness) on a three-level scale (Pass / Minor / Fail).

Contribution and verification. The evaluation rubric, sampling strategy, prompt design, and statistical analysis (Wilson confidence intervals) were all designed by the author. The model's only role was applying the rubric to each extracted section. All Fail verdicts were manually spot-checked against the original SEC filings and confirmed to be correct. Results are reported in Table B.1.

Table B.1: LLM-assisted extraction validation results ($n = 200$ sections, 100 per section type).

Criterion	Strict	%	Lenient	%	95 % CI (lenient)
Correct section	192/200	96.0	192/200	96.0	[92.3 – 98.0 %]
Start boundary	191/200	95.5	193/200	96.5	[93.0 – 98.3 %]
End boundary	192/200	96.0	193/200	96.5	[93.0 – 98.3 %]
Content quality	191/200	95.5	195/200	97.5	[94.3 – 98.9 %]
Completeness	189/200	94.5	191/200	95.5	[91.7 – 97.6 %]
Overall (all five)	179/200	89.5	187/200	93.5	[89.2 – 96.2 %]

Code Development

Tool. AI-powered code completion (integrated in VS Code).

Purpose. Code completion during the development of the data-pipeline scripts that build the panel and extract the 10-K text sections.

Contribution and verification. All generated code was reviewed, tested, and modified by the author. The overall architecture, variable definitions, and analytical choices are the author's own.

Writing Assistance

Tool. AI-powered writing assistant (integrated in VS Code).

Purpose. Grammar and style suggestions for English-language prose in the thesis text.

Contribution and verification. All substantive content, argumentation, and interpretation are the author's own. The tool was used only for language polishing; every suggestion was reviewed before acceptance.